LF^*RF_*Q = Q FOR Q SUPPORTED ON F^{-1}Z WHEN F IS AN ISOMORPHISM OVER Z AND FLAT ALONG IT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1993-2000 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 2002-2043, the complete author proof (40 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 14 of queue_spaces-pushouts.json (pb=1808, stmt 1993-2000, proof 2002-2043).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0AET: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 1957–1967. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AEQ: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 1840–1850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05E9: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2043–2043. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : Y \to X$ be a morphism of algebraic spaces over S. Let $Z \subset X$ be a closed subspace. Assume $f^{-1}Z \to Z$ is an isomorphism and that f is flat in every point of $f^{-1}Z$. For any Q in $D_{QCoh}(\mathcal{O}_Y)$ supported on $|f^{-1}Z|$ we have $Lf^*Rf_*Q = Q$.*

Proof. We show the canonical map $Lf^*Rf_*Q \to Q$ is an isomorphism by checking on stalks at $\overline{y}$. If $\overline{y}$ is not in $f^{-1}Z$, then both sides are zero and the result is true.

Date: stacks-project:spaces-pushouts.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Assume the image $\overline{x}$ of $\overline{y}$ is in Z. By Lemma 0AEQ we have $Rf_*Q_{\overline{x}} = Q_{\overline{y}}$ and since f is flat at $\overline{y}$ we see that

$$(Lf^*Rf_*Q)_{\overline{y}} = (Rf_*Q)_{\overline{x}} \otimes_{\mathcal{O}_{X,\overline{x}}} \mathcal{O}_{Y,\overline{y}} = Q_{\overline{y}} \otimes_{\mathcal{O}_{X,\overline{x}}} \mathcal{O}_{Y,\overline{y}}$$

Thus we have to check that the canonical map

$$Q_{\overline{y}} \otimes_{\mathcal{O}_{X,\overline{x}}} \mathcal{O}_{Y,\overline{y}} \longrightarrow Q_{\overline{y}}$$

is an isomorphism in the derived category. Let $I_{\overline{x}} \subset \mathcal{O}_{X,\overline{x}}$ be the stalk of the ideal sheaf defining Z. Since $Z \to X$ is locally of finite presentation this ideal is finitely generated and the cohomology groups of $Q_{\overline{y}}$ are $I_{\overline{y}} = I_{\overline{x}}\mathcal{O}_{Y,\overline{y}}$-power torsion by Lemma 0AET applied to Q on Y. It follows that they are also $I_{\overline{x}}$-power torsion. The ring map $\mathcal{O}_{X,\overline{x}} \to \mathcal{O}_{Y,\overline{y}}$ is flat and induces an isomorphism after dividing by $I_{\overline{x}}$ and $I_{\overline{y}}$ because we assumed that $f^{-1}Z \to Z$ is an isomorphism. Hence we see that the cohomology modules of $Q_{\overline{y}} \otimes_{\mathcal{O}_{X,\overline{x}}} \mathcal{O}_{Y,\overline{y}}$ are equal to the cohomology modules of $Q_{\overline{y}}$ by More on Algebra, Lemma 05E9 which finishes the proof. $\square$